



U.S. Department
of Transportation
**Pipeline and
Hazardous Materials
Safety Administration**

400 Seventh Street, S.W.
Washington, D.C. 20590

DOT-SP 8865
(TENTH REVISION)

EXPIRATION DATE: December 31, 2009

(FOR RENEWAL, SEE 49 CFR § 107.109)

1. GRANTEE: Carleton Technologies Inc.
Orchard Park, NY
2. PURPOSE AND LIMITATIONS:
 - a. This special permit authorizes the manufacture, mark, sale and use of a gas storage system consisting of a non-DOT specification cylinder equipped with pyrotechnic relief devices. The cylinder must conform with all regulations applicable to a DOT Specification 39 cylinder for the transportation in commerce of helium. This special permit provides no relief from the Hazardous Materials Regulations (HMR) other than as specifically stated herein.
 - b. The safety analyses performed in development of this special permit only considered the hazards and risks associated with transportation in commerce. The safety analyses did not consider the hazards and risks associated with use as a component of a transport vehicle or other device, or other uses not associated with transportation in commerce.
3. REGULATORY SYSTEM AFFECTED: 49 CFR Parts 106, 107 and 171-180.
4. REGULATIONS FROM WHICH EXEMPTED: 49 CFR §§ 173.302a(a) and 175.3 in that the use of a non-DOT specification package is not authorized, except as specified herein.
5. BASIS: This special permit is based on the application of Carleton Technologies Inc. dated January 1, 2006, submitted in accordance with § 107.109.

6. HAZARDOUS MATERIALS (49 CFR § 172.101):

Proper Shipping Name/ Hazardous Material Description	Hazard Class/ Division	Identi- fication Number	Packing Group
Helium, compressed	2.2	UN1046	N/A

Note: A missile gas storage system or component which contains a quantity of pyrotechnic materials must be classed and approved as provided for in § 173.56. If the pyrotechnic material augments the volume of the gas in the cylinder, or in any way enhances the performance of the compressed gas, the device must be tested in the same configuration as when shipped.

7. SAFETY CONTROL MEASURES:

a. PACKAGING - Prescribed packaging is described as a gas storage system and consists of a non-DOT specification cylinder equipped with two (2) Division 1.4 pyrotechnic devices. The gas storage system must conform with Carleton Technologies Inc.'s drawing 3505-0002, P/N B11461-1, B13748-1, and B18755-1 dated March 29, 1990, and other referenced drawings on file with the Office of Hazardous Materials Special Permits and Approvals (OHMSPA) and DOT Specification 39 (§ 178.65), except as follows:

§ 178.65(a) *Type, size, service pressure and test pressure.*

Each cylinder must be made from steel and have a seamless cylindrical body with the bottom attached by means of a circumferential mechanical joint with a service pressure not to exceed 8,500 psig.

(1) *Size limitation.* Maximum water capacity may not exceed 170 cubic inches.

(2) *Test pressure.* The minimum test pressure is 15,000 psig.

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§ 178.65(b) *Material*. Material must be one of the following heat treatable steels: Carpenter custom 455 or PH 13-8 Mo steels.

§ 178.65(c) *Manufacture*.

* * *

(4) Add - "All welds must be 100 percent radiographically inspected".

§ 178.65(f) *Pressure tests*.

(1) * * *

(2) One cylinder randomly selected from each lot must be hydrostatically tested to failure. The entire lot must be rejected if (see § 178.65(h)) a failure occurs at a gauge pressure less than 2.5 times the marked service pressure.

(3) A "lot" is defined as the quantity of pressure vessels fabricated from the same heat of steel, manufactured by the same process and heat treated in the same equipment under same conditions of time, temperature and atmosphere and shall not exceed a quantity of 200.

§ 178.65(g) *Flattening test*.

* * *

(3) (Add) Flattening between flat plates is authorized. Flattening must continue until test specimen fails.

(4) Cylinder or test ring may not crack when flattened so that their outer edges are not more than 15 times wall thickness apart. Failure point must be recorded in the inspector's test report as multiples of the cylinder wall thickness.

§ 178.65(i) *Markings.*

* * *

(2) Required markings are as follows:

(i) "DOT-SP 8865" instead of "DOT-39".

(ii) ("NRC" Does not apply).

(iii), (iv), (v), (vi), (vii) * * *

(viii) ("Refilling penalty" does not apply).

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b. TESTING - Prototype packagings must have been qualified under the qualification program as described in the application on file prior to initial shipment.

c. OPERATIONAL CONTROLS -

(1) The cylinder must be shipped in strong outside packagings in accordance with the provisions of § 173.301(g) (9).

(2) Each cylinder may be refilled a maximum of five times to a pressure not exceeding 8,500 psig, provided each cylinder is visually inspected and proof pressure tested to 15,000 psig prior to refilling. Refilling is authorized to be performed by Carleton Technologies only. Topping off is considered a refilling. The serial number of each cylinder that has been refilled must be reported to the AAHMS at the times of the next renewal application.

8. SPECIAL PROVISIONS:

a. In accordance with the provisions of Paragraph (b) of § 173.22a, persons may use the packaging authorized by this special permit for the transportation of the hazardous materials specified in paragraph 6, only in conformance with the terms of this special permit.

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- b. A person who is not a holder of this special permit, but receives a package covered by this special permit, may reoffer it for transportation provided no modification or change is made to the package or its contents and it is offered for transportation in conformance with this special permit and the HMR.
- c. A current copy of this special permit must be maintained at each facility where the package is offered or reoffered for transportation.
- d. Each packaging manufactured under the authority of this special permit must be either (1) marked with the name of the manufacturer and location (city and state) of the facility at which it is manufactured or (2) marked with a registration symbol designated by the Office of Hazardous Materials Special Permits and Approvals for a specific manufacturing facility.
- e. A current copy of this special permit must be maintained at each facility where the package is manufactured under this special permit. It must be made available to a DOT representative upon request.
- f. Each cylinder design must be qualified by subjecting a representative cylinder to a pressure reversal (cycle) test from approximately zero to service pressure at a rate not to exceed 10 cycles per minute. Test cylinder must withstand at least 1,000 cycles without distortion or defect.
- g. Packagings permanently marked 'DOT-E 8865', prior to October 1, 2007 may continue to be used under this special permit for the remaining service life of the packaging or until the special permit is no longer valid. Packagings marked on or after October 1, 2007 must be marked 'DOT-SP 8865'.
- h. Shipping papers displaying 'DOT-E 8865' may continue to be used until October 1, 2007, provided the special permit remains valid.
- 9. MODES OF TRANSPORTATION AUTHORIZED: Motor vehicle, rail freight, cargo aircraft only.
- 10. MODAL REQUIREMENTS: A current copy of this special permit must be carried aboard aircraft used to transport packages

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covered by this special permit. The shipper shall furnish a current copy of this special permit to the air carrier before or at the time the shipment is tendered.

11. COMPLIANCE: Failure by a person to comply with any of the following may result in suspension or revocation of this special permit and penalties prescribed by the Federal hazardous materials transportation law, 49 U.S.C. 5101 et seq:

- o All terms and conditions prescribed in this special permit and the Hazardous Materials Regulations, Parts 171-180.
- o Persons operating under the terms of this special permit must comply with the security plan requirement in Subpart I of Part 172 of the HMR, when applicable.
- o Registration required by § 107.601 et seq., when applicable.

Each "Hazmat employee", as defined in § 171.8, who performs a function subject to this special permit must receive training on the requirements and conditions of this special permit in addition to the training required by §§ 172.700 through 172.704.

No person may use or apply this special permit, including display of its number, when the special permit has expired or is otherwise no longer in effect.

Under Title VII of the Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (SAFETEA-LU)- 'The Hazardous Materials Safety and Security Reauthorization Act of 2005' (Pub. L. 109-59), 119 Stat. 1144 (August 10, 2005), amended the Federal hazardous materials transportation law by changing the term 'exemption' to 'special permit' and authorizes a special permit to be granted up to two years for new special permits and up to four years for renewals.

12. REPORTING REQUIREMENTS: Shipments or operations conducted under this exemption are subject to the Hazardous Materials Incident Reporting requirements specified in 49 CFR §§ 171.15 - Immediate notice of certain hazardous materials incidents, and 171.16 - Detailed hazardous materials incident reports. In addition, the grantee(s) of this exemption must notify the Associate Administrator for

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Hazardous Materials Safety, in writing, of any incident involving a package, shipment or operation conducted under terms of this exemption.

Issued in Washington, D.C.

A handwritten signature in blue ink, appearing to read "R. Ryan Porter". The signature is fluid and cursive, with a large loop at the end.

for Robert A. McGuire
Associate Administrator
for Hazardous Materials Safety

Address all inquiries to: Associate Administrator for Hazardous Materials Safety, Pipeline and Hazardous Materials Safety Administration, Department of Transportation, Washington, D.C. 20590. Attention: PHH-31.

Copies of this special permit may be obtained by accessing the Hazardous Materials Safety Homepage at
http://hazmat.dot.gov/sp_app/special_permits/spec_perm_index.htm

Photo reproductions and legible reductions of this special permit are permitted. Any alteration of this special permit is prohibited.

PO: CWF/AM